

PID Controllers

Industrial Control Systems · SCADA Hub

COURSE TOPICS

1. **Introduction:** Feedback control concept and the uniform mathematical structure underlying temperature, pressure, flow, and level applications.
2. **Control Loop Fundamentals:** Process variable, setpoint, error, and manipulated variable — vocabulary prerequisite for PID algorithm study.
3. **PID Algorithm:** Proportional, integral, and derivative terms; the specific problem each term solves; combined three-term controller behavior.
4. **Tuning Methods:** Ziegler-Nichols closed-loop and open-loop, lambda tuning, IMC; method selection criteria and failure modes.
5. **Process Characterization:** First-order plus dead time model, step test procedure, and extracting gain, time constant, and dead time from equipment.
6. **Cascade Control:** Inner and outer loop design, commissioning sequence, and disturbance rejection improvement over single-loop control.
7. **Digital Implementation:** Sampling rate, velocity-form vs. position-form algorithm, bumpless transfer, and anti-windup on saturated outputs.
8. **PLC Implementation:** PID function block configuration across Rockwell, Siemens, Beckhoff, and Modicon; scaling conventions; commissioning procedure.
9. **Troubleshooting:** Oscillation, offset, slow response, and derivative chatter — each symptom mapped to a specific tuning adjustment.
10. **Practice Lab:** Flow loop step-test tuning, cascade temperature/flow loop, anti-windup configuration, cold-start commissioning sequence.

LEARNING OUTCOMES

- Perform a step test on live equipment and extract first-order plus dead time model parameters.
- Apply Ziegler-Nichols and lambda tuning methods to self-regulating process loops.
- Design and commission a cascade control loop with sequenced inner and outer tuning.
- Configure anti-windup and bumpless transfer on a PLC PID function block.
- Diagnose oscillation, steady-state offset, and derivative chatter from trend data alone.

Certification Relevance

- **ISA CCST Level III** — Advanced process control and loop tuning are tested at the senior technician level.
- **ISA CAP** — Certified Automation Professional exam covers control system design and PID fundamentals explicitly.
- **PE Exam (Control Systems)** — Professional Engineer licensing tests control theory, system response, and stability analysis.

Next Steps

1. Take PID alongside IEC 61131-3. PLC PID blocks make sense once you understand both the algorithm and the language.
2. Use Scilab (free) or MATLAB (student license) to simulate step tests before commissioning on live equipment.
3. ISA-5.1 (Instrumentation Symbols) and ISA-18.2 (Alarm Management) are the surrounding standards worth reviewing.
4. ISA CCST is the right entry credential, with ISA CAP as the next step for advanced process control work.